

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 2

Year: 2019

Subject: 340152 Electrical Measuring Instruments

Section: 5-6

Date: 24 January 2020

Time: 08.00-10.00

Name _____ ID: _____ Class _____

Directions: The test is designed to measure your comprehension. The test is divided into 13 questions. There will be 9 pages (including this page) and they are worth 70 points.

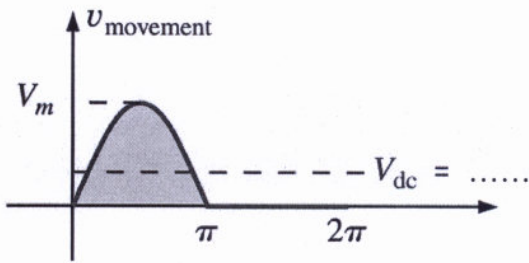
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Q1) Find the V_{Dc} (V_{Ave}) of a Half-wave rectified waveform by using Integration.

(5 marks)



Integration Formulas:

$$\int 1 dx = x + c$$

$$\int \sin x dx = -\cos x + c$$

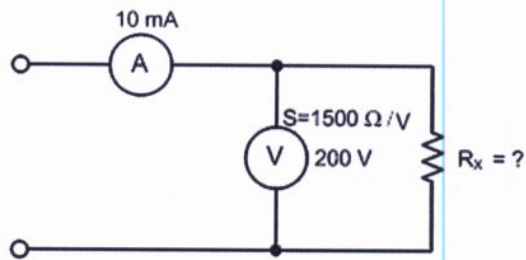
$$\int \cos x dx = \sin x + c$$

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Q2) A voltmeter having a sensitivity of $1,500 \Omega/V$, reads 200 V on its 300 V scale when connected across an unknown resistor (R_x) in series with an ammeter and the ammeter reads 10 mA, Calculate the following items:

- (a) The apparent resistance of the unknown resistor (R_T),
- (b) The actual resistance of the unknown resistor (R_x),
- (c) % error due to the loading effect.

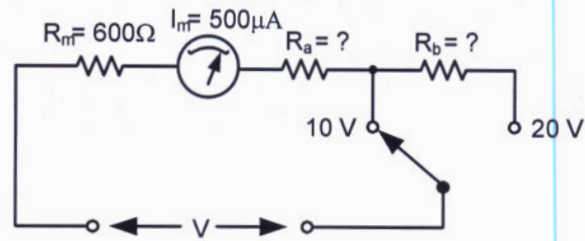
(10 marks)



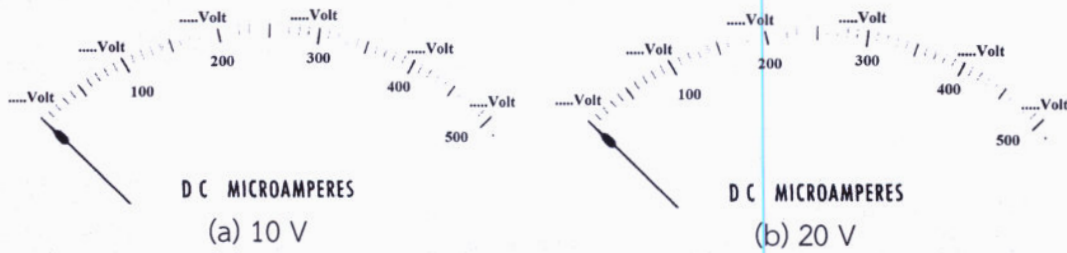
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Q3) A PMMC instrument with FSD = $500\ \mu\text{A}$ and $R_m = 600\ \Omega$, is to be employed as a volt-meter with ranges of 10 V, and 20 V. (10 marks)

(a) Calculate the required values of multiplier resistors for the circuit.



(b) Determine how the voltmeter scale should be marked.

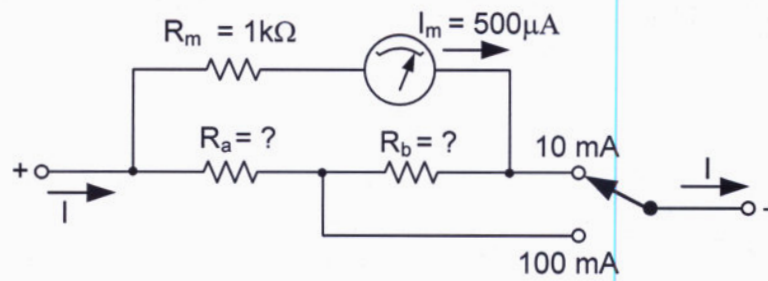


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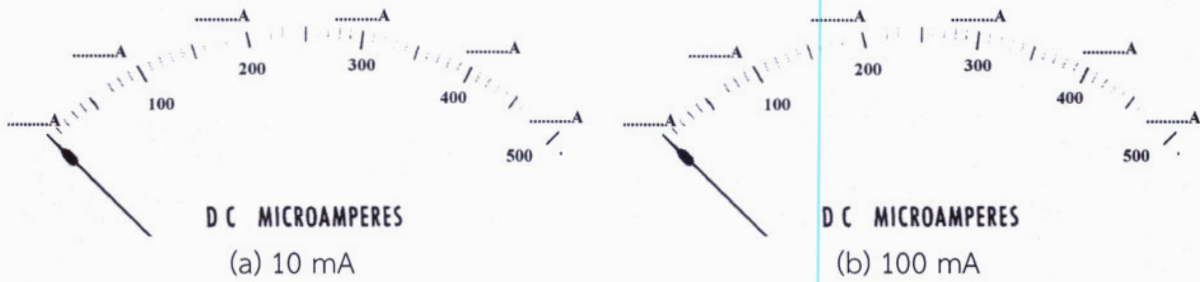
Q4) A PMMC instrument has $500\ \mu\text{A}$ and a coil resistance of $1\text{k}\Omega$. (10 marks)

(a) Calculate the required resistance values to convert the instrument into an ammeter (multi-range) with an FSD of (a) 10 mA and (b) 100 mA .

(10 marks)



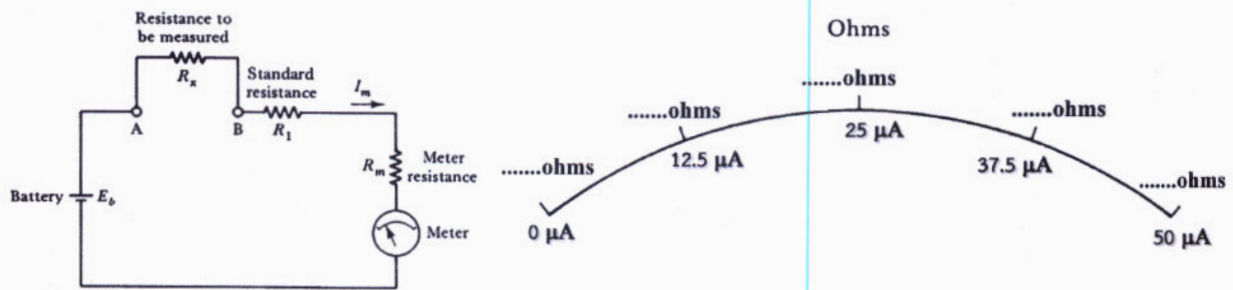
(b) Determine how the Ammeter scale should be marked.



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Q5) The series ohmmeter in figure is made up of a 1.5 V battery, a $50\ \mu\text{A}$ meter, $R_m = 600\ \Omega$.

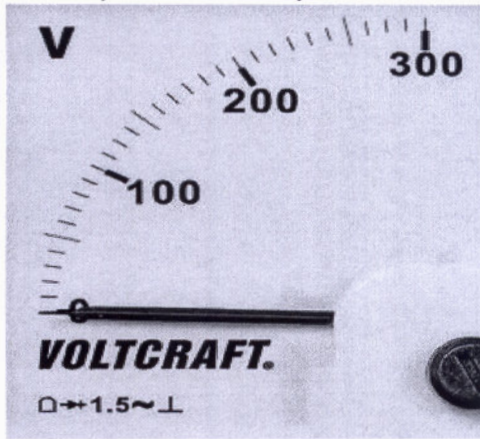
- (a) Calculate the value of R_1 to make the current to be full scale.
 (b) Determine how the resistance scale should be marked at $R_x = 0$ and $R_x = \infty$.
 (c) Determine how the resistance scale should be marked at 0.75 FSD, 0.5 FSD, and 0.25 FSD. (10 marks)



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Q6) Identify each of the symbols illustrated in Figure.

(3 marks)



a) $\square$:

b) 1.5 :

c) $\perp$:

Q7) Define the following terms:

(4 marks)

(a) Accuracy :

(b) Precision :

(c) Measurement :

(d) Instrument :

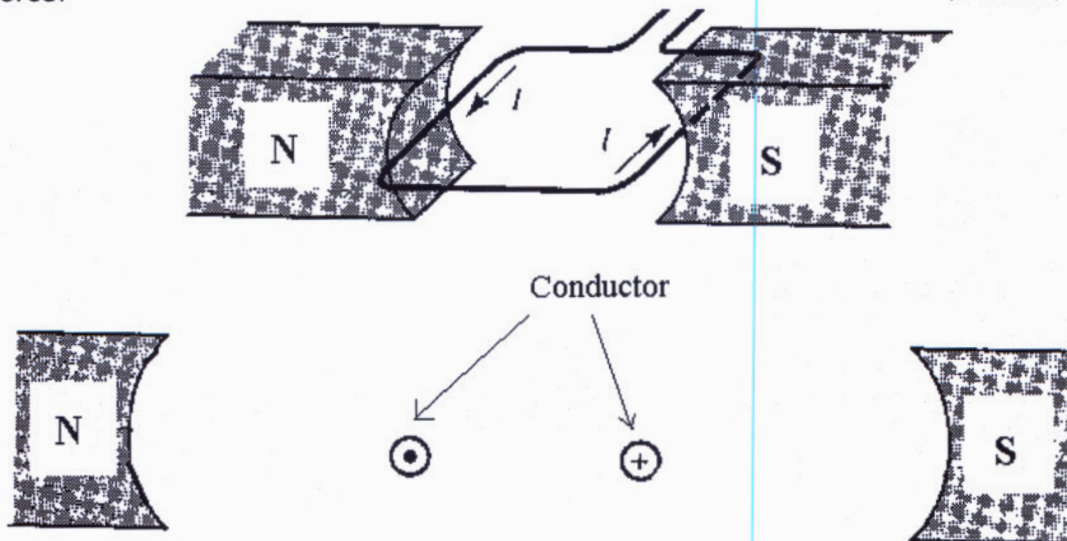
Q8) State the three major categories of error.

(3 marks)

.....

Q9) Using *Right Hand Grip Rule* to sketch the direction of the **magnetic field** and **force**.

(3 marks)



Q10) List and explain the **three forces** involving in the moving system of a deflection instrument. (3 marks)

[illegible]

Q11) Describe the basic concept of meter movement. (2 marks)

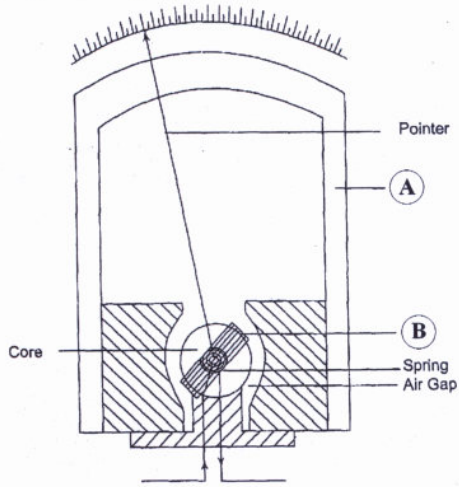
[illegible]

Q12) The expected value of a resistor is $10\text{ k}\Omega$, but the measurement using ohmmeter yields a value of $10.34\text{ k}\Omega$. Calculate (a) absolute error, (b) % error, (c) accuracy, and (d) % of accuracy. (5 marks)

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Q13) Identify each part of the instrument:

(2 marks)



Part A is

Part B is.....

Assoc. Prof. Dr. Pichet Sriyanyong

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